Course milestone M1: Curiosity Committed and the Research Problem

HONR 46400 · Evidence-Driven Research

What to submit

One file, `lastname_m01.pdf`. Everything this milestone asks for goes inside it, as sections in the order below.

Submit it on Brightspace, under **Assignments** then **M01**. Brightspace carries the deadline.

Your work lives in a Colab notebook, which has no export-to-PDF button. **Making the PDF you hand in**, at the end of this document, is the one-minute routine that turns the notebook into the file you upload.

This milestone presents **Book Milestone 1: Your curiosity, committed** (checkpoint, version 1) of the course book, EDR|AI.

Open the milestone workbook in Colab

This chapter closes Studio 1: Begin with your curiosity. Its lessons each ended at **It is your turn**; what you wrote there arrives here as working material, and this is where the pieces become one artifact you can defend.

What this milestone produces. Your committed curiosity with its starting belief and the evidence that would revise it, the brainstormed candidates with what you kept and rejected and why, and one chosen research problem, scored and red-teamed, with your expected answer on record.

What you bring

Check you are carrying each of these before you start. If one is missing, go back and work that lesson's **It is your turn** first; the practice below assumes it.

- Lesson 1: From Curiosity to a Research Problem: your committed curiosity, the brainstormed candidates with NEW elements marked and kept or rejected with reasons, your chosen research problem scored and red-teamed, and your expected answer.

The practice

1. Copy your opening move's four lines as the artifact's version zero, dated.
2. Widen the curiosity: write two candidate directions yourself, then run the structured brainstorm and mark everything it introduced as NEW.
3. Choose the research problem and score it honestly on importance, feasibility, and contribution, one sentence each.
4. Write your expected answer and name one real person or role for whom a different answer would be genuinely surprising.
5. Keep the trail: candidates kept and rejected, each with its reason, and the NEW elements you adopted as your own.

A version, not a pass

Your milestone artifact is a dated, numbered **version** with the reason for the version attached. When later evidence changes it, you write the next version rather than editing the last one, because the sequence of changes is itself part of your research record.

The four rails, here

Four concerns cross every studio in this book. At this milestone they take these specific forms.

- **Ethics, permissions, and data exposure.** Note now whether chasing this curiosity would require people, and flag it before anything is designed.
- **Evidence, provenance, and reproducibility.** Your starting belief is a claim to check, not to defend; write it so evidence can move it.

- **AI activity, verification, and human decisions.** Brainstorm freely, and mark every AI stretch NEW; start a simple exchange log now, and Studio 2 turns it into your AI Research Ledger.
- **Uncertainty, claim boundary, and revision history.** Write what would change your mind while nothing is yet at stake; that sentence is the seed of every later boundary.

Where this milestone sits

This is the first of the book's twelve milestones. Milestone 2 sets the rules that protect this curiosity, opens the ledger that records every delegation, verifies your first AI assistance, and closes the arc by declaring your formal research question. The chain ends at Milestone 12: Your release and next cycle, where you decide whether your finished research artifact leaves your hands.

What sends you back

Studio 3′s evidence map and Studio 4′s diagnosis both send curiosity back here. A revised curiosity is progress on the record, not a restart.

How this milestone is assessed

Each row scores **0**, **1**, or **2**. **10 points total.**

#	Criterion	0	1	2
1	The milestone artifact exists in full, specific to your own project	absent, or generic text that would fit any project	present but incomplete, or specific in some parts and generic in others	complete and unmistakably about your project
2	The milestone is written as a dated, numbered version with its reason	no version, or a version with no reason given	versioned, but the reason is decorative rather than an account of what changed	versioned with a reason a reader could use to reconstruct your thinking
3	Every judgment in the artifact is defended by you, not asserted by a tool	conclusions appear with no reasoning, or reproduce a tool's wording	reasoning present but thin where the decision was hardest	each judgment carries a reason you could defend under questioning
4	The four rails are carried in the form this studio requires	one or more rails absent where this studio required them	all rails present but one is nominal	every rail addressed in the form this studio requires
5	The milestone's own product is present and defensible: Your committed curiosity with its starting belief and revising evidence, the brainstormed candidates with NEW elements marked and kept or rejected with reasons, and one chosen research problem, scored and red-teamed, with your expected answer on record	the milestone's own product is missing	present but would not survive a careful question	present and defensible as stated

Your workbook

Open the workbook with the link above. It walks these steps with a cell for each one, ends by writing your milestone version with its reason, and adds the rows this studio contributes to your exchange log and your Research Dossier.

Making the PDF you hand in

Colab has no export-to-PDF button. You make the PDF by **printing the notebook to a file**, which takes about a minute once you know the preparation steps. Skipping them is the most common way a milestone arrives looking half finished, because a notebook prints what is on the screen and nothing else.

1. Prepare the notebook

1. **Run the whole thing.** `Runtime` → `Run all`, then wait for it to finish. A cell you never ran prints with nothing underneath it, so an analysis you really did can reach the grader as an empty box.
2. **Expand every collapsed section.** Anything folded shut prints as a heading with nothing beneath it. Open all of them, including any Colab folded for you.
3. **Shorten any output that scrolls.** When Colab puts a scrollbar on an output, or tells you the output is truncated, only the visible part prints. Show the first twenty rows instead of the whole table, or summarise it, then rerun that cell. A reader did not want the whole table anyway.
4. **Read it once, top to bottom.** You are about to freeze it.

2. Print it to a file

`File` → `Print` opens your browser's print dialog. Set the destination to **Save as PDF**, open the dialog's further settings, and turn on **Background graphics**, so code cells keep their shading and your figures keep their backgrounds. Save it under the file name this milestone asks for.

Wide tables and wide figures are the usual casualty. If something runs off the right edge, switch the layout to **landscape** or reduce the scale, and print again.

3. Check the PDF before you upload it

Open the file you just made and confirm three things: every figure is whole rather than sliced by a page break, no output stops mid-line, and the pages run in order with nothing missing. A PDF is an artifact like any other in this course, so you verify it before your name goes on it.

One route not to take

Ask an AI assistant how to export a Colab notebook and it will very likely hand you `jupyter nbconvert --to pdf`. Do not spend your afternoon on it. Inside Colab that route needs a large LaTeX installation, runs for several minutes, and then either fails or quietly drops the symbols and emoji these notebooks are full of. Printing to a file is the route that works.

That is a small, cheap instance of the rule this whole course runs on. The confident answer is not automatically the correct one, and the way you find out is to check it against what actually happens.